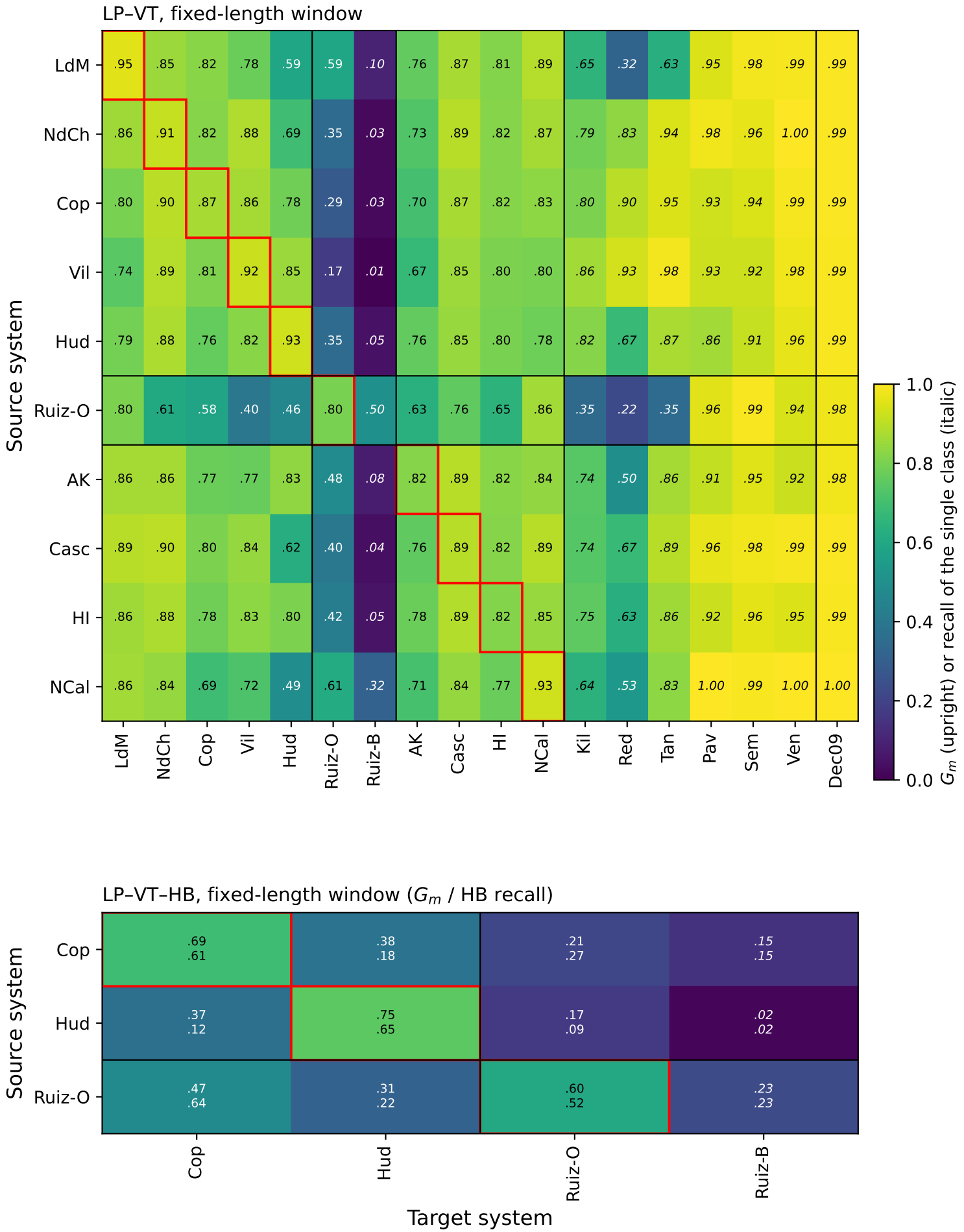Transfer matrices, gradient-boosted trees, source scaling; red: within-system validation